



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0353



Agricultural Machinery – Walking-Type Agricultural Tractor WESPRO WAE-MT800

Test Report Number	2023 – 0353
Date of Issuance	October 30, 2024
Valid Until	October 30, 2029
Agricultural Machinery	Walking-Type Agricultural Tractor
Type	Rotary tilling, Axle-driven
Brand	WESPRO
Model	WAE-MT800
Test Requested by	WESPRO INDUSTRIAL TRADING 114 Magsaysay St., 6 th Avenue, East Grace Park, Caloocan City

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 346:2022	AMTEC Test
Power requirement of the prime mover, kW, maximum	13.5	5.60 ^a
Actual Field Capacity, ha/h	ND ^b	0.174
Peak Transmission Efficiency, %, minimum	80	NM ^c
Breakdowns/malfunctions during the five-hour continuous running test.	Shall have none	NM ^c

^a As per the engine nameplate.

^b Not specified by the requesting party.

^c Laboratory test was not conducted as decided by the requesting party.

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly and installation	Provided for handling only
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	WESPRO INDUSTRIAL TRADING
4. Prime mover used	5.60 kW KRESS KE207 Gasoline Engine
5. Serial number	800AE2308001MT
6. Number of operators	One
7. Handling and stability of the WTAT	The machine was stable during operation.
8. Manipulating of the operating levers	The main clutch lever and gear shift lever are accessible to the left-hand side, and of the operator.
9. Replacing and adjusting of parts	Locally parts can be replaced. Engine placement can be adjusted through its engine base.
10. Safety features	Transmission system cover and mud guard were provided.
11. Failure or abnormalities that may be observed on the WTAT or its component parts.	No failure or abnormalities were observed on the machine or its component parts during and after the test operation. Accumulation of weeds on the rotary tilling blades was observed during the test operation (see Figure 5).
12. The WTAT shall be generally made of steel bars and sheet metals.	The machine is generally made of round bars and metal sheet.
13. The chain and sprocket used shall conform to PAES 303:2000 (Engineering materials – Roller chains and sprockets for agricultural machines – Specifications and applications) and shall have provisions for refilling of lubricating oil.	There is a provision for refilling of lubrication oil for the chain and sprocket transmission system. However, the chain and sprocket assembly were not accessible thus conformity to PAES 303:2000 was not verified.

OBSERVATIONS (Cont'n)

14. The hand grip shall be made of non-slip material and shall have a minimum nominal diameter of 25 mm.	The hand grip is made of rubber and has a nominal diameter 25.11 mm.
15. Mechanism for handle bar height adjustment with at least three settings shall be provided. The handle bar height shall be 650 mm to 800 mm.	The handle bar height can be adjusted through a lever. The operating height is measured from the ground at 1140 mm.
16. The hitch of the WTAT shall be in accordance with the specifications of PAES 107:2000 (Agricultural machinery - Hitch for walking-type agricultural tractor – Specifications).	The WTAT has a one-hole hitch with a diameter of 18.38 mm. See Specifications table for the complete dimensions of one-hole hitch
17. The speed control lever shall be provided and accessible to and operated from the perspective of the operator. It shall be provided with a detent mechanism capable of setting and retaining/locking the speed at any point within the range of the engine speed.	The speed control lever is provided and is located at the right-hand side of the operator. Detent mechanism was not provided.
18. The main clutch lever shall be provided and accessible to and operated from the perspective of the operator.	The main clutch lever is provided and is located on the left-hand side of the operator.
19. Stand assembly for use when the WTAT is not in operation shall be provided.	Stand assembly was provided and is located at the rear of the machine.
20. There shall be earmuffs or other ear protection device provided for the operator to use when 95 dB(A) is exceeded during operation. Recommendations for use of ear protection device shall likewise be indicated in the operator's manual.	The noise level measured during the test was 84.1 dB(A). No earmuffs or other protection devices were provided as it did not exceed 95 dB(A).
21. The WTAT shall be free from any manufacturing defects that may be detrimental to its operation.	The WTAT is free from any manufacturing defects.
22. All metal surfaces shall be painted properly.	All metal surfaces are painted properly.
23. Parts of the WTAT that are exposed to the operator shall be free from sharp edges and rough surfaces.	The WTAT is free from sharp edges or rough surfaces.
24. Appropriate labels, safety symbols, and warning notices shall be provided in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	Labels are provided on the WTAT. Warning notices and safety symbols are provided on the prime mover of the WTAT.

OBSERVATIONS (Cont'n)

25. All controls shall be in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	The controls are labeled, accessible, and can be easily maneuvered by the operator under normal working conditions.
26. All exposed, hot surfaces shall be provided with heat shield or protection to minimize the possibility of inadvertent contact with the operator.	The exhaust outlet of the prime mover was directed away from the operator. However, heat shield or protection was not provided.
27. Mechanism for emergency stop of the engine and immediate load disengagement of the transmission system shall be provided.	Load may be disengaged by releasing the idler pulley through the main clutch lever.
28. Mechanism for belt transmission adjustment shall be provided.	Belt tension adjustment is provided through adjustment of engine along its base.
29. Belt cover and mud guard shall be provided. All other moving parts shall have safety guards. When the PTO is in use, a cover or casing that protects the sides of the PTO shaft shall be fitted. An additional non-rotating casing shall also be provided when the PTO is not in use. This casing shall enclose the PTO shaft completely and shall be fixed to the tractor body.	Belt cover and mudguard are provided.
30. There shall be a provision for means to minimize vibration.	It was not verified if the machine's vibration was minimized/reduced due to limited technical capabilities to quantify and compare vibration levels during the test operation.
31. The hexagonal axle and shall be in accordance with the specifications of PAES 108:2000 (Agricultural machinery – Hexagonal axle and hub for walking-type agricultural tractor – Specifications).	The axle is 250 mm in length with 23.50 mm width across the flat. See Specifications table (page 7) for all the dimensions.
32. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	July 12, 2023
Location of Test	College, Los Baños, Laguna 14°09'54.18 N, 121°15'06.73" E
Items Tested	Operator's Manual, Depth of Rototilling, Width of Rototilling, Field Capacity, Field Efficiency, Noise Level, Number of Operators, and Fuel Consumption Rate

METHODS OF TEST

The rotary tiller was tested based on the PNS/BAFS 348:2022 Agricultural Machinery – Walking-type Agricultural Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

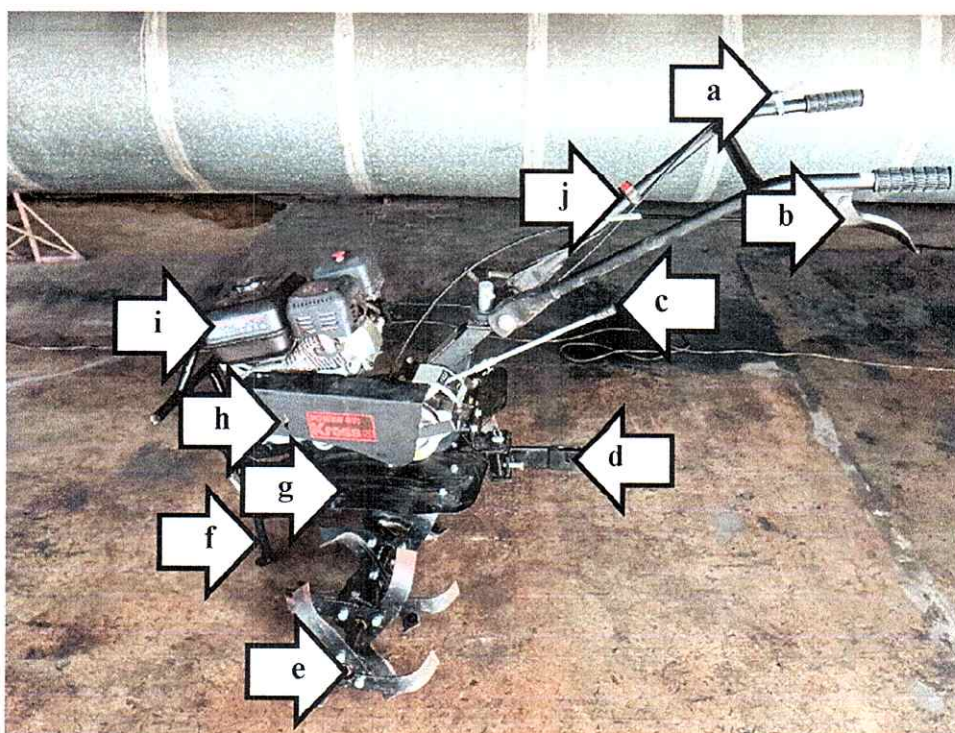


Figure 1. Parts of the WESPRO WAE-MT800 Walking-Type Agricultural Tractor powered by 5.60 kW KRESS KE207 Gasoline Engine include the following: (a) Speed control lever, (b) Main clutch lever, (c) Gear shift lever, (d) Drawbar, (e) Rotary tilling blades, (f) Stand, (g) Mud guard, (h) Power transmission system with cover, (i) Prime mover, and (f) Emergency stop button.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1570	1390
1.1.2	Width	560	940
1.1.3	Height	1090	1140
1.1.4	Ground clearance	ND	100
1.2	Weight, kg	ND	66
2	Engine		
2.1	Brand	KRESS	KRESS
2.2	Model	KE207	KE207
2.3	Make or manufacturer	China	China
2.4	Serial number	2021020149374	2021020149374
2.5	Type	ND	Four-stroke, single inclined cylinder
2.6	Rated power, kW	5.59 ^a	5.59 ^a
2.7	Rated speed, rpm	3600	3600 ^a
2.8	Displacement, cm ³	212	212 ^a
2.9	Fuel type	Gasoline	Gasoline
2.10	Tank capacity, L	3.6	3.52
2.11	Cooling system	ND	Air-cooled
2.12	Starting system	Handling clutch	Rope-recoil
3	Type of clutch		
3.1	Main clutch	ND	Idler
3.2	Steering clutch	NA	NA
3.3	Tilling clutch	ND	Gear shift lever
4	Power transmission system		
4.1	Engine to input	ND	V-belt and Pulley
4.2	Engine ^b	ND	19.13 × 2B × 89
4.3	Input ^b	ND	16.78 × 2B × 123
4.4	Belt size	ND	ND ^b

^a The data were obtained from the components' nameplates and specifications sheet provided by the requesting party.

^b Not specified on the components' markings.

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
5	Axle		
5.1	Type	ND	Hexagonal
5.2	Width across the flat, mm	ND	23.50
5.3	Diameter, mm	ND	24.88
5.4	Overall length of shaft, mm	ND	250
6	Hitch point		
6.1	Type	ND	One-hole
6.2	Dimensions		
6.2.1	H	ND	64.38
6.2.2	h	ND	11.05
6.2.3	b	ND	8.39
6.2.4	c	ND	21.85
6.2.5	l	ND	91.06
6.2.6	d _i	ND	18.38
6.2.7	d _o	ND	NA
6.2.8	t	ND	5.34
6.2.9	r	ND	5.22
7	Rotary tilling wheel		
7.1	Number of blades	24	3-3-3-3 3-3-3-3
7.2	Number of flanges	ND	8
7.3	Dimension, L × D, mm	NA	940 × 29.70
7.4	Blade thickness, mm	2	2.69
7.5	Blade shape	ND	J-shaped
7.6	Operating width, mm	910	940
8	Safety devices	ND	Emergency stop switch, belt cover, and mud guard
9	Special features	ND	None

^a The data were obtained from the specifications sheet by the requesting party.

Note: NA – Not Applicable, ND – No Data

RESULTS

Item		Data
1	Test conditions	
1.1	Condition of field	
1.1.1	Type	Upland
1.1.2	Location	College, Los Baños, Laguna
1.1.3	Soil type	Lipa loam
1.1.4	Dimensions of field	
1.1.4.1	Length, m	32.0
1.1.4.2	Width, m	16.0
1.1.4.3	Area, m ²	512
1.1.5	Soil moisture content, % _{d.b.}	8.05
1.1.6	Soil resistance, kg/cm ²	10
1.1.7	Spacing of stubbles, rows × hills, mm	NA
1.1.8	Height of stubbles, mm	NA
1.1.9	Weed density	Medium
1.2	Condition of the tractor	
1.2.1	Implement/Attachment used	
1.2.1.1	Type	Rotary tiller
1.2.1.2	Operating width, mm	865

Note: NA – Not Applicable

RESULTS (Cont'n)

Item		Data
2	Field performance	
2.1	Type of field operation	Primary tillage
2.2	Field operational pattern	Continuous pattern with rounded corners
2.3	Traveling speed, kph	2.13
2.4	Average depth of rotary tilling, mm	99
2.5	Average width of rotary tilling, mm	766
2.6	Actual field capacity	
2.6.1	ha/h	0.174
2.6.2	h/ha	5.78
2.7	Theoretical field capacity	
2.7.1	ha/h	0.185
2.7.2	h/ha	5.42
2.8	Field efficiency, %	93.93
2.9	Fuel consumption	
2.9.1	L/h	0.967
2.9.2	L/ha	5.65
2.10	Noise at operator's ear level, dB(A)	84.1
2.11	Overlapped, %	24.34

OBSERVATIONS

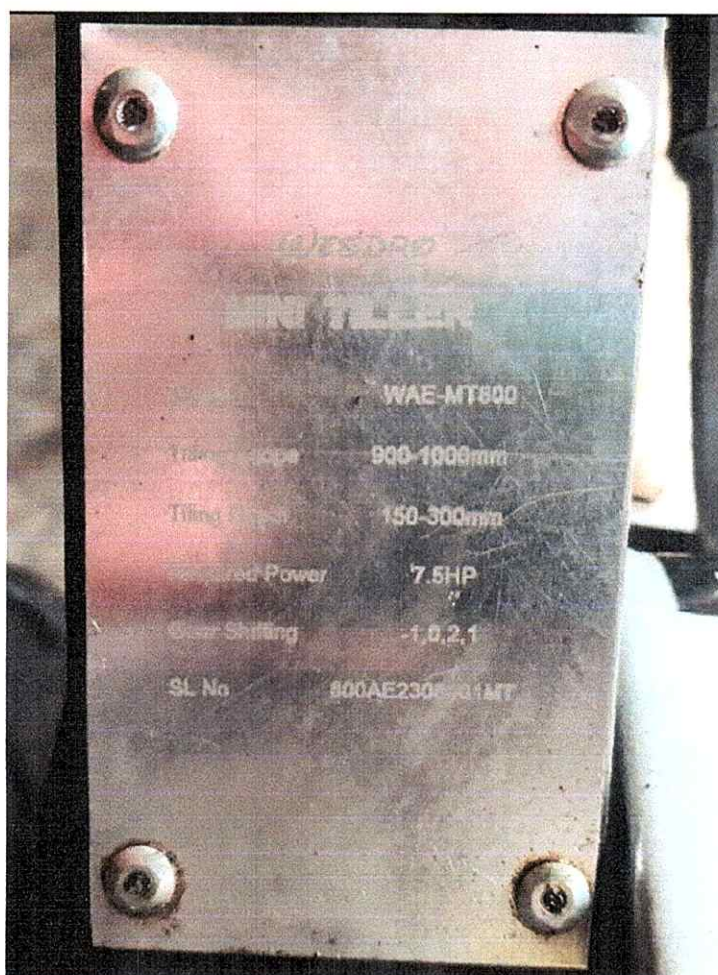


Figure 2. Nameplate of the WESPRO WAE-MT800 Walking-Type Agricultural Tractor.

OBSERVATIONS (Cont'n)

Figure 3. Nameplate of the prime mover, and markings of the WESPRO WAE-MT800 Walking-Type Agricultural Tractor.



Figure 4. Clogging of weeds on the rotary tilling blades of WESPRO WAE-MT800 Walking-Type Agricultural Tractor.

OBSERVATIONS (Cont'n)

Figure 5. Condition of field before the operation.



Figure 6. Quality of work.

Tested by:



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Test Engineer



RAXL P. AGAPITO

Test Engineer

Verified by:



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Approved for Release:



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07 NOV 2024

Date

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